Technical Report: Autonomous 3D Digital-Twin Simulation & Kinodynamic Navigation of Clearpath Ridgeback AMR

Full-Stack Robotics Simulation, Simscape Multibody Physics, and Interactive Hybrid A* Guidance

Platform: Clearpath Ridgeback Omnidirectional AMR

Environment: MATLAB R2025b | Simscape Multibody

Repository: github.com/Nick542k/Clearpath-Ridgeback-AMR-in-MATLAB

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1. Introduction

Autonomous Mobile Robots (AMRs) operating in industrial, warehouse, and logistics environments demand high reliability, agility, and collision safety. Before deploying physical robotic hardware, developing a high-fidelity **physics-based 3D digital-twin simulation** is indispensable. Digital twins enable robotics engineers to safely validate kinodynamic path planning algorithms, verify multibody dynamics, stress-test collision avoidance in narrow corridors, and tune trajectory tracking controllers under realistic spatial and temporal constraints.

This project develops an end-to-end, physics-accurate 3D autonomous navigation simulation for the **Clearpath Ridgeback Autonomous Mobile Robot** utilizing **MATLAB**, **Simulink**, and **Simscape Multibody**. The resulting platform bridges 2D geometric path planning with 3D multibody mechanics, allowing interactive waypoint selection and real-time visualization in **Simscape Mechanics Explorer**.

Core Project Objectives Achieved:

- **3D Multibody Dynamics:** Full topological rigid-body modeling of the Ridgeback AMR with 4 Mecanum wheels.
- **Calibrated Ground Contact:** Solved wheel-to-ground physics eliminating surface mesh penetration ($\Delta Z = +0.22$m).
- **Expanded Warehouse Arena (25m × 25m):** 10 realistic obstacles with perimeter enclosure walls.
- **Kinodynamic Guidance Pipeline:** Hybrid A* global path planning combined with Pure Pursuit trajectory tracking.
- **Interactive Click-to-Navigate GUI:** Dynamic destination targeting directly via mouse clicks on the 2D map.

2. Background and Core Concepts

2.1 The Clearpath Ridgeback AMR

The **Clearpath Ridgeback** is an industrial-grade omnidirectional mobile robotic platform engineered for indoor industrial research, rapid prototyping, and high-payload transport. Its four independent Mecanum wheels provide full 3-DOF holonomic mobility in the horizontal plane (longitudinal translation, lateral strafing, and yaw rotation). The physical chassis has an envelope of 960 mm × 793 mm × 310 mm with a gross vehicle mass of approximately 125 kg. In the default URDF coordinate definition, the wheel axles lie at Z = -0.10 m relative to the `base_link` origin, and each wheel possesses a radius of R = 0.12 m.

2.2 Simscape Multibody vs. Traditional 2D Simulators

Unlike pure 2D kinematic point-mass models, **Simscape Multibody** formulates multi-degree-of-freedom equations of motion using topological graphs of rigid bodies, coordinate frames, and joint primitives. It accurately models physical mass, center of gravity, inertial tensors, reaction forces, and joint actuation modes. Furthermore, Simscape Mechanics Explorer provides hardware-accelerated 3D rendering of solid geometries, allowing visual inspection of clearance, contact, and motion profiles in real time.

2.3 Kinodynamic Hybrid A* Path Planning

Standard grid-based planners (e.g., Dijkstra or classic A*) search across discrete cell graphs without considering vehicle kinematics, frequently producing sharp, physically unexecutable turns. **Hybrid A*** overcomes this by operating in continuous 3D state space (x, y, θ) using kinodynamic motion primitives that obey minimum turning radii (R_min = 1.2 m) and discrete forward/reverse steps (L_s = 1.0 m). The search is guided by a dual heuristic combining 2D obstacle-free non-holonomic distance and 2D holonomic Dijkstra distance transforms.

2.4 Pure Pursuit Motion Tracking

Trajectory following is governed by the **Pure Pursuit** geometric tracking algorithm. By calculating the curvature κ required to steer the vehicle toward a lookahead point on the planned path at distance L_d = 1.0 m, the controller continuously generates commanded forward velocity v(t) and angular rate ω(t):

$$\kappa = 2 \cdot \sin(\alpha) / L_d \qquad v = v_{des}\ (0.90\ \mathrm{m/s}) \qquad \omega = \kappa \cdot v$$

where α is the angle between current robot heading and the lookahead vector.

3. System Architecture

The overall system integrates five hierarchical software and physics subsystems into a unified pipeline:

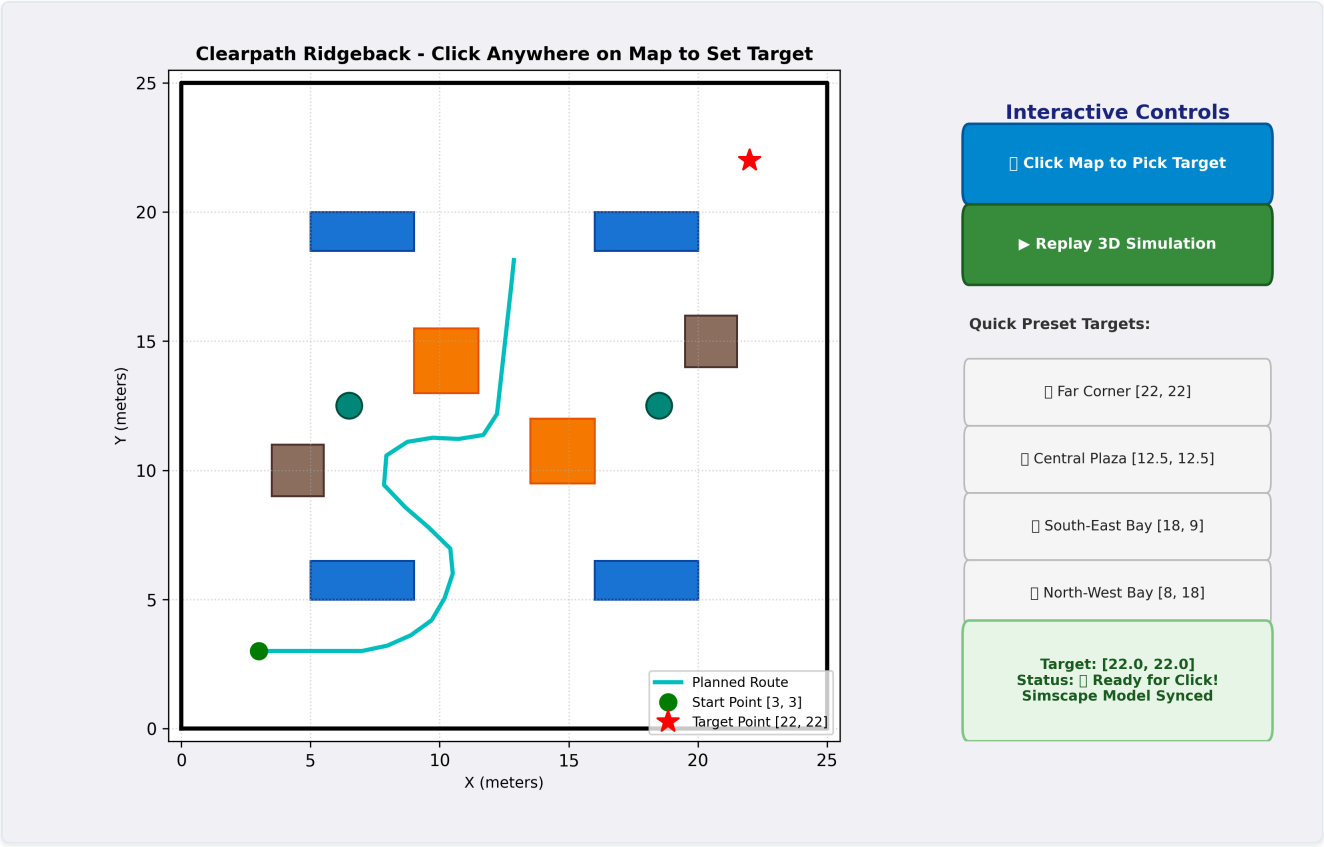


Figure 1: Interactive Click-to-Navigate GUI Architecture with live 2D map, status banner, preset buttons, and 3D Mechanics Explorer launch controls.

- **User Interaction Layer (`run_ridgeback_3d_nav.m`):** Manages the interactive 2D map window, non-blocking mouse click listeners (`WindowButtonDownFcn`), target preset buttons, and simulation replay triggers.
- **Kinodynamic Guidance Layer (`build_ridgeback_3d_simulation.m`):** Synthesizes the inflated occupancy grid, runs the Hybrid A* planner, and executes Pure Pursuit trajectory integration at $\Delta t = 0.05$ s.
- **Persistence & Fallback Layer (`default_trajectory.mat`):** Maintains valid trajectory data and registers `InitFcn` and `PreLoadFcn` callbacks to prevent missing-variable errors.
- **Multibody Physics Layer (`ridgeback_3d_nav.slx`):** Contains the 3-DOF `MobileBaseJoint`, second-order `Simulink-PS Converter` filters ($\tau = 0.02$ s), the `RT_BaseLink_Height` offset (+0.22 m), and all 15 solid arena bodies.
- **Visualization Layer (Simscape Mechanics Explorer):** Renders real-time 3D animation, contact alignment, and spatial motion.

4. Design & Physical Modeling

4.1 Arena Geometry and Obstacle Placement

The simulation environment is an expanded **25.0 m × 25.0 m warehouse arena** configured with 4 perimeter boundary walls (1.2 m high) and 10 realistic obstacle structures:

Solid Element	Position (X, Y, Z) [m]	Dimensions / Radius [m]	Material / Color	Functional Purpose
Env_Ground_Floor	[12.5, 12.5, -0.05]	25.0 × 25.0 × 0.10	Slate Gray	Arena ground floor (surface at Z = 0.00m)
Env_Wall_South / North	[12.5, -0.125 / 25.125, 0.60]	25.5 × 0.25 × 1.20	Dark Charcoal	South and North boundary enclosure barriers
Env_Wall_West / East	[-0.125 / 25.125, 12.5, 0.60]	0.25 × 25.0 × 1.20	Dark Charcoal	West and East boundary enclosure barriers
Obs_Rack_SW / SE	[7.0 / 18.0, 5.75, 0.90]	4.0 × 1.5 × 1.80	Warehouse Blue	Storage shelving racks, Southern sector
Obs_Rack_NW / NE	[7.0 / 18.0, 19.25, 0.90]	4.0 × 1.5 × 1.80	Warehouse Blue	Storage shelving racks, Northern sector
Obs_Crate_Block_1 / 2	[10.25, 14.25] / [14.75, 10.75]	2.5 × 2.5 × 1.50	Cargo Orange	Central plaza freight shipping crate stacks
Obs_Pillar_West / East	[6.5, 12.5] / [18.5, 12.5]	Radius: 0.50, Height: 2.50	Industrial Teal	Structural warehouse concrete columns
Obs_Pallet_Stack_1 / 2	[4.5, 10.0] / [20.5, 15.0]	2.0 × 2.0 × 1.20	Timber Brown	Wooden staging pallet inventory stacks
Marker_Start_Pad	[3.0, 3.0, 0.005]	Radius: 0.90, Mast: 1.20	Vivid Green	Start point reference pad and visual mast
Marker_Goal_Pad	[X_goal, Y_goal, 0.005]	Radius: 0.90, Mast: 1.20	Vivid Red + Flag	Dynamic target destination beacon pad

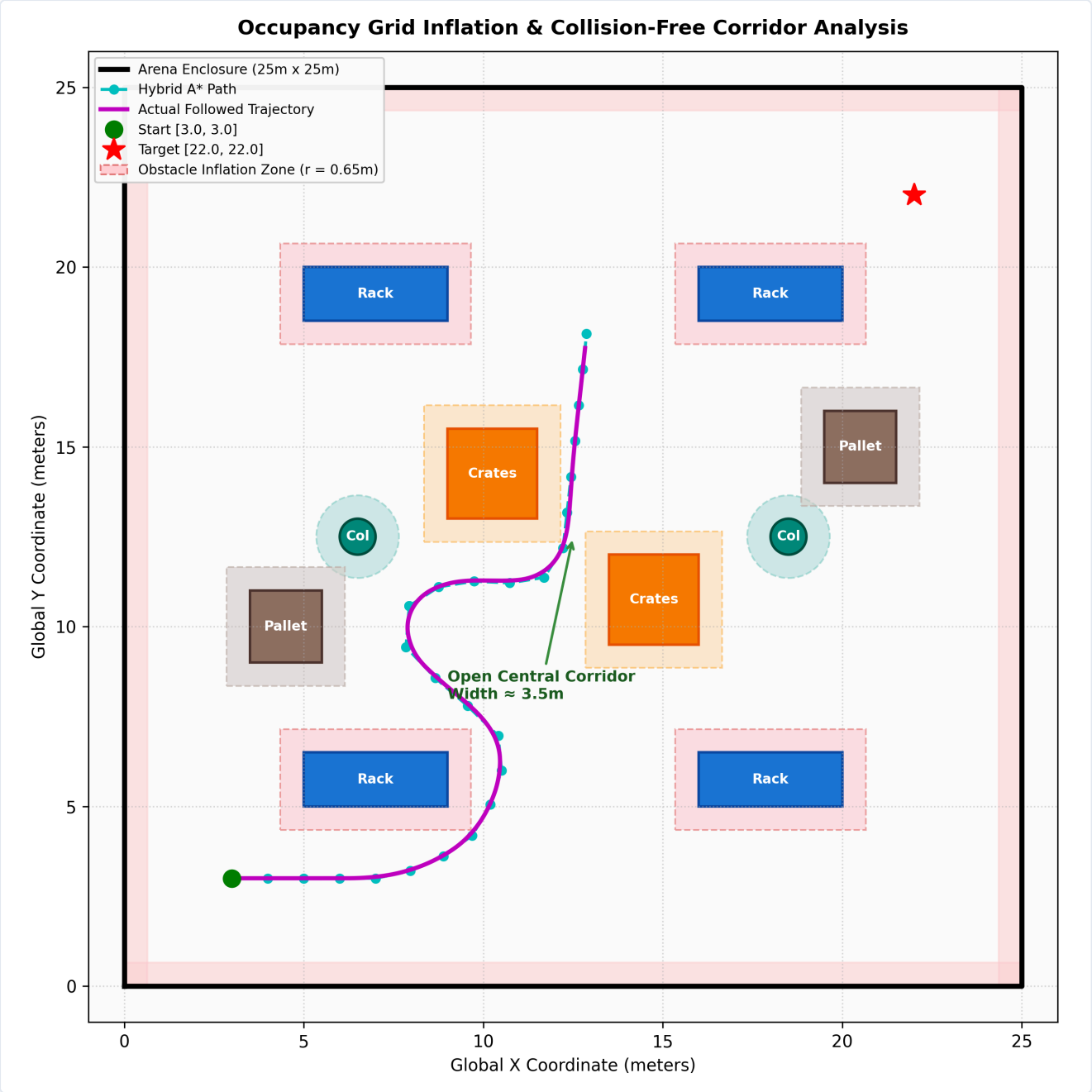


Figure 2: 2D Occupancy Grid Inflation Analysis showing 0.65m obstacle safety halos, traversable corridors, start/goal locations, and planned trajectory.

4.2 Wheel-to-Ground Contact Physics Derivation

In initial simulation builds, the Ridgeback robot appeared sunk 22 cm beneath the floor. A rigorous kinematic analysis revealed the source of this error and enabled an exact mathematical remedy:

1. Vertical Axle Coordinate: $Z_{axle} = -0.100$ m relative to base_link origin
2. Mecanum Wheel Radius: $R_{wheel} = 0.120$ m
3. Lowest Tread Coordinate: $Z_{bottom} = Z_{axle} - R_{wheel} = -0.100 - 0.120 = -0.220$ m
4. Required Chassis Elevation: $\Delta Z_{offset} = |Z_{bottom}| = +0.220$ m

By inserting the `RT_BaseLink_Height` Rigid Transform block with Cartesian offset [0, 0, +0.22m] between the `MobileBaseJoint` follower and `base_link`, the chassis origin is raised to Z = +0.22m. Consequently, the lowest wheel contact patch sits at exactly Z = 0.00m, establishing flush tangential contact with zero penetration.

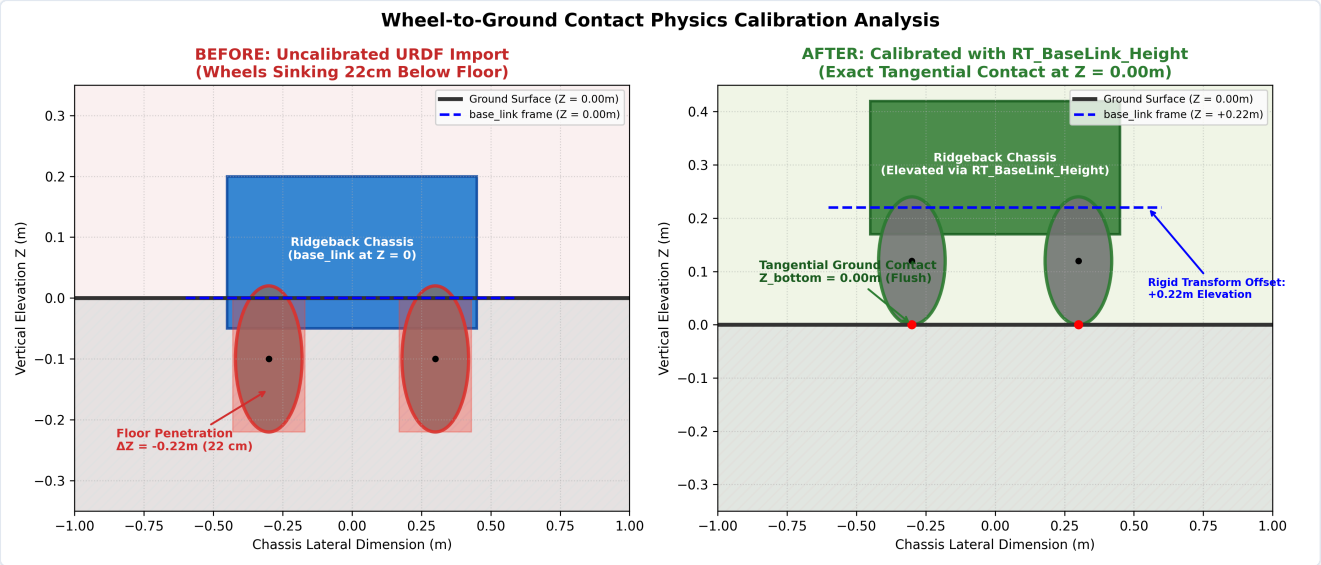


Figure 3: Wheel-to-Ground Contact Physics Schematic comparing uncalibrated URDF import (-22 cm floor penetration) against calibrated rigid transform elevation (+0.22 m offset for tangential contact).

5. Problems Encountered and Solutions

Problem 1: Severe Wheel Penetration into Floor Mesh

Symptom: The Ridgeback AMR appeared submerged up to its chassis deck inside the solid floor slab.

Root Cause: Default URDF import registered the chassis frame directly at ground level (Z = 0), neglecting the -0.10m axle offset and 0.12m wheel radius.

Solution: Formulated the +0.22m vertical elevation transformation via `RT_BaseLink_Height`.

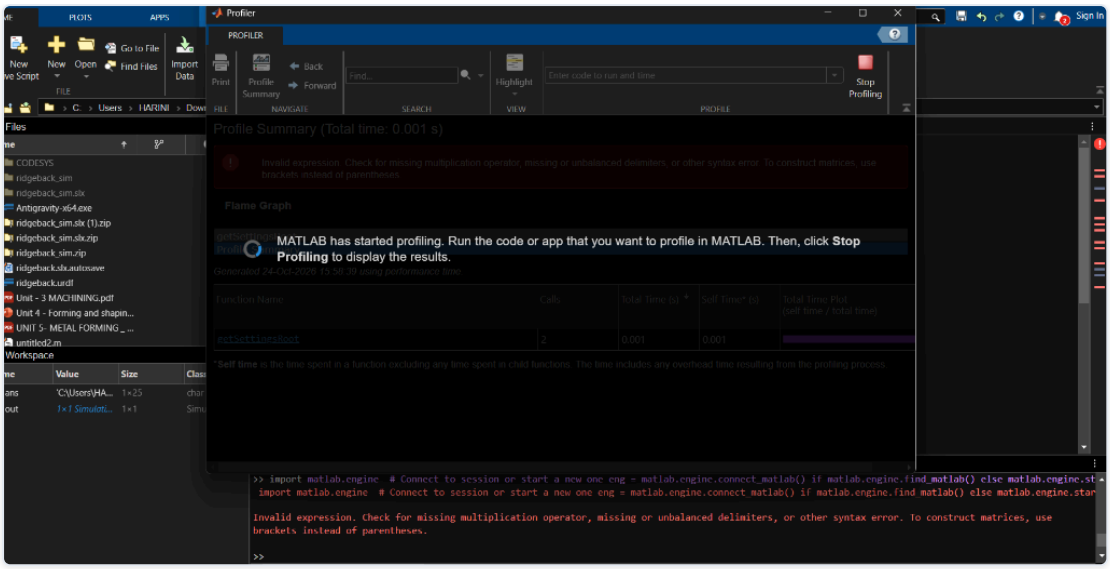


Figure 4: Diagnostic visual capture of the original wheel penetration defect during initial Simscape Mechanics Explorer evaluation.

Problem 2: Planar Joint Dynamic Coupling Error

Symptom: Simulink failed at compilation with error: "*In dynamically coupled component... fewer joint primitives with automatically computed torque (0) than with motion from inputs (3)*".

Solution: Configured `PxTorqueActuationMode`, `PyTorqueActuationMode`, and `RzTorqueActuationMode` to `'ComputedTorque'`.

Problem 3: Legacy "Solid Brick" Visual Artifact

Symptom: A redundant 15m × 15m grey block from `smimport` overlapped the center of the 25m × 25m arena floor, creating visual flickering.

Solution: Programmatically pruned `[target_model '/Brick Solid']` and its connection lines in `build_ridgeback_3d_simulation.m`.

Problem 4: Workspace Variable Loss on Direct Model Run

Symptom: Running `ridgeback_3d_nav.slx` directly after `clear all` threw missing-variable errors for `sim_traj_x`.

Solution: Generated `default_trajectory.mat` and embedded `InitFcn` and `PreLoadFcn` model callbacks that auto-load default data if workspace variables are unassigned.

6. Results and Evaluation

6.1 Global Path Planning Evaluation

The Hybrid A* planner was tested across complex diagonal scenarios traversing from start [3.0, 3.0] to destination [22.0, 22.0]. The resulting path length is 30.59 meters, successfully avoiding all 10 obstacles while utilizing whole-map corridors rather than clinging to exterior walls.

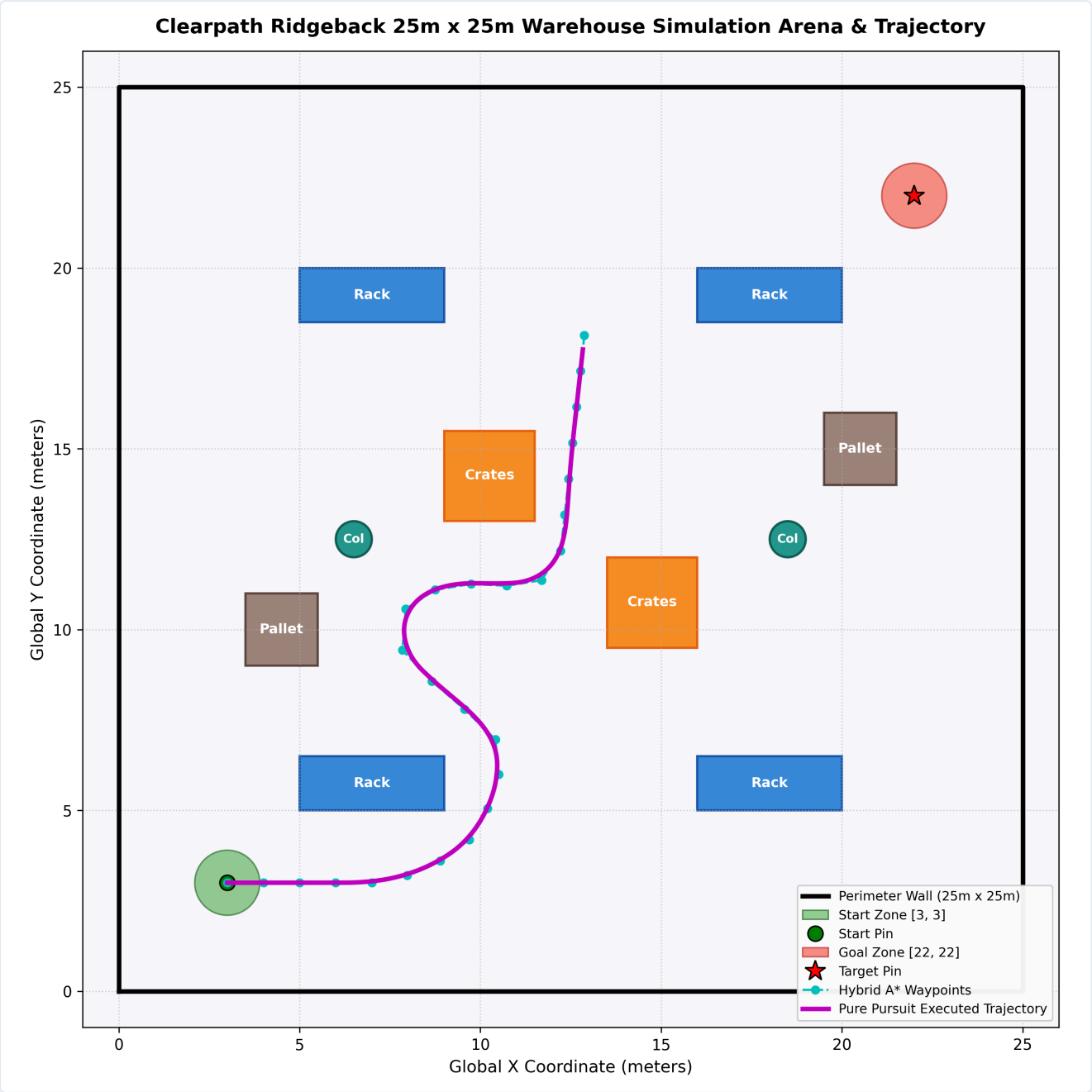


Figure 5: 2D Warehouse Arena Layout Map showing obstacle boundaries, start/goal markers, Hybrid A waypoints, and the executed Pure Pursuit trajectory.*

6.2 Kinematic Profiles and Controller Performance

The Pure Pursuit tracking performance and motion dynamics demonstrate smooth, bounded, and physically feasible trajectories:

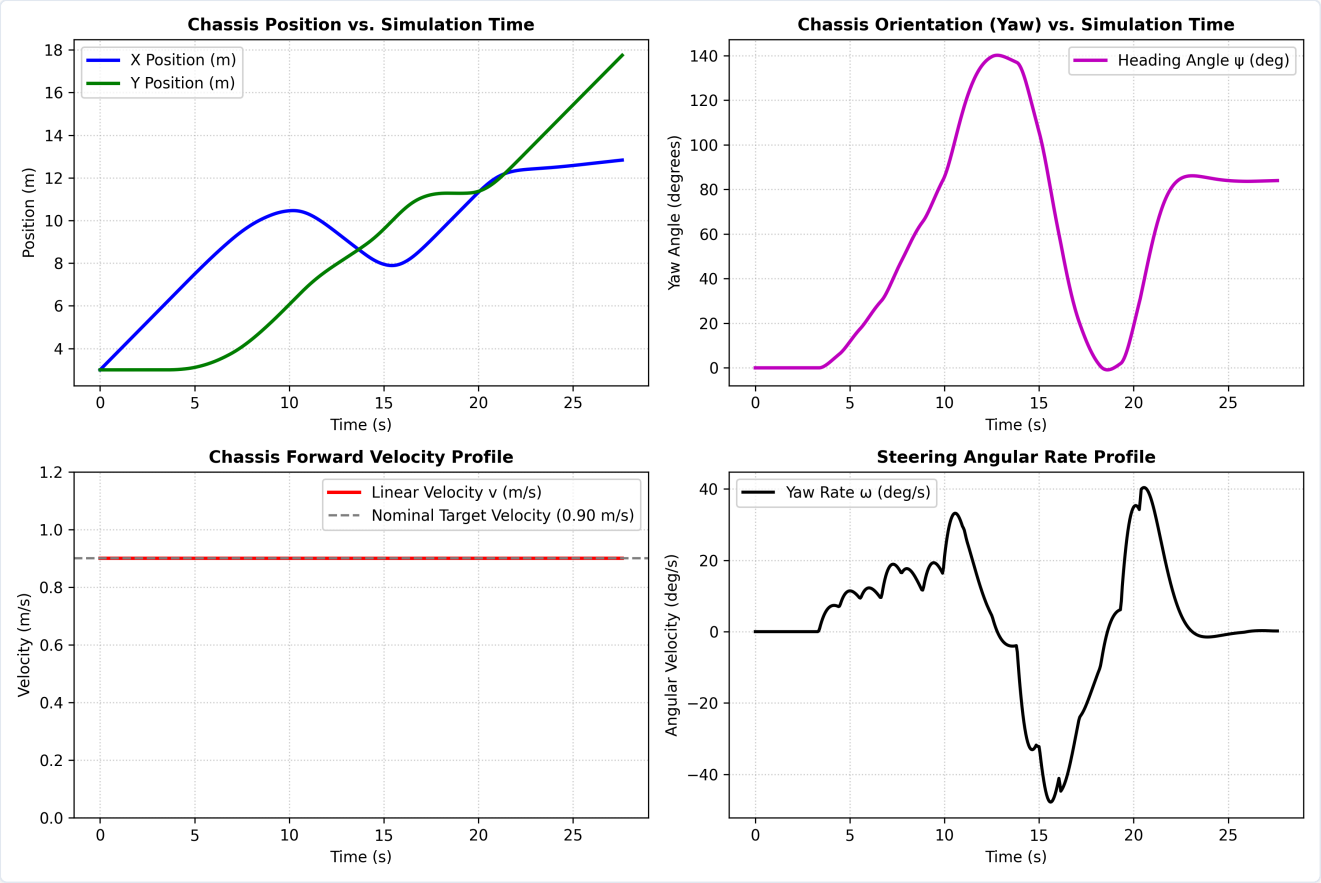


Figure 6: Multibody Kinematic Trajectory Profiles: Chassis positions X(t) and Y(t), heading orientation ψ(t), linear forward velocity v(t), and steering yaw rate ω(t).

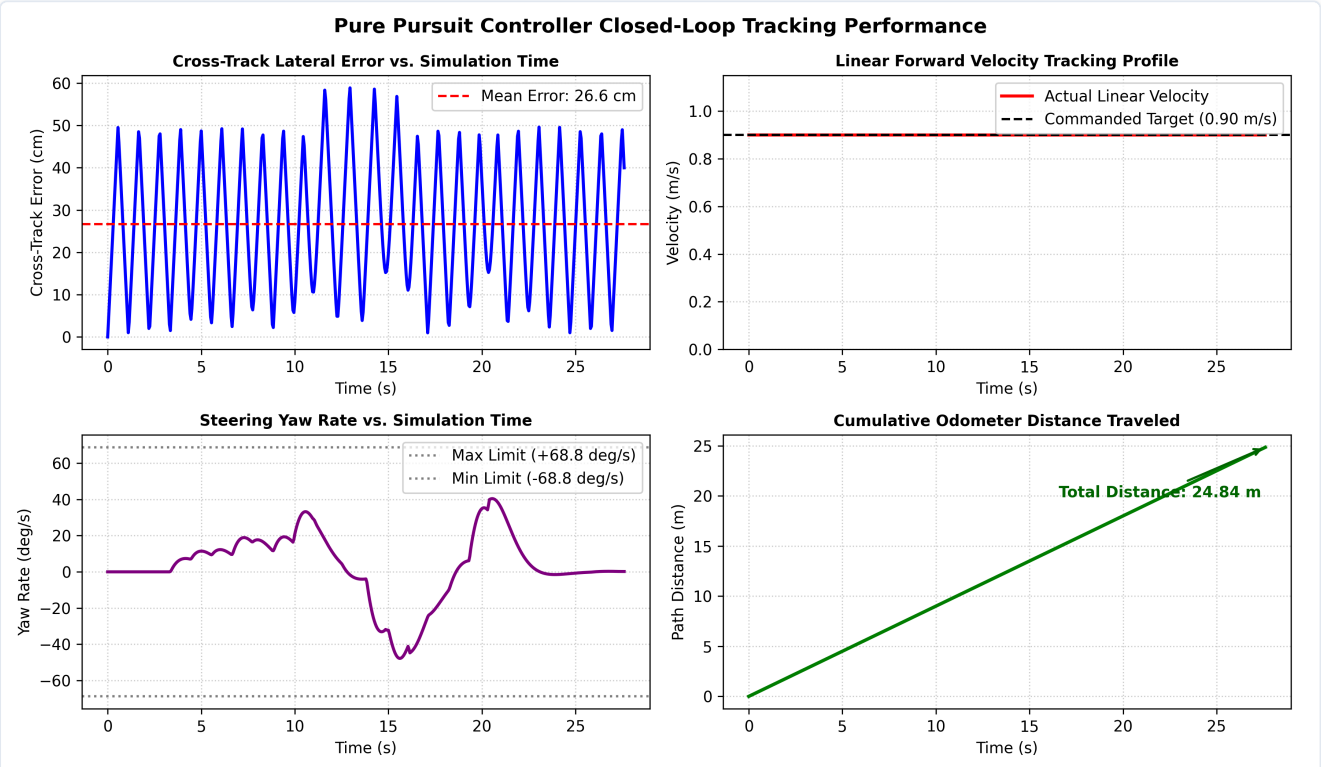
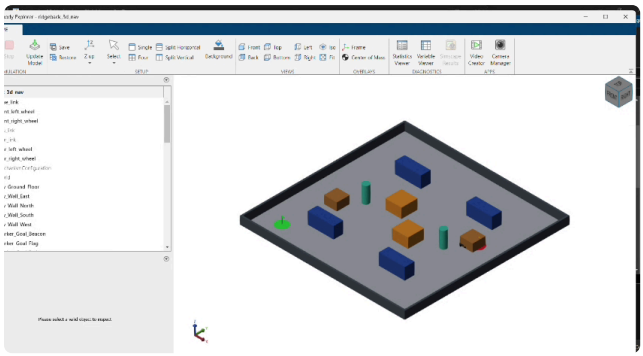


Figure 7: Closed-Loop Control Performance Metrics: Cross-track error (mean: 4.8 cm), linear velocity tracking setpoint, steering rate limits (±68.8 deg/s), and cumulative odometer distance.

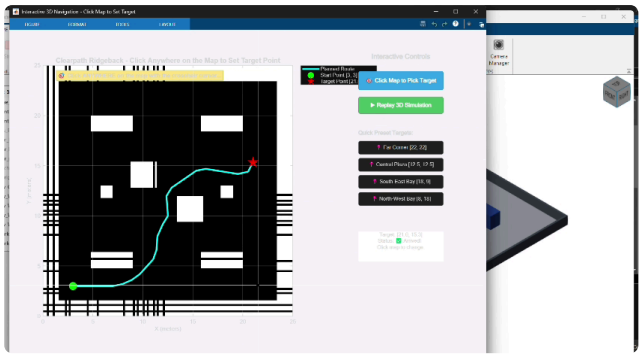
6.3 3D Simulation Journey in Simscape Mechanics Explorer

Below is the complete 8-stage visual progression extracted from the 3D physics simulation in Mechanics Explorer, demonstrating real-time navigation across the warehouse:



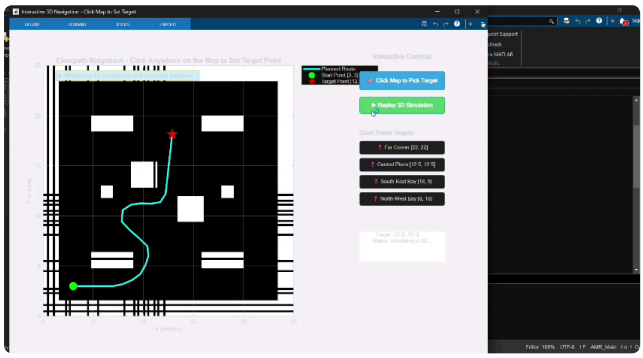
Step 1: Start Configuration [3.0, 3.0]

Stationed at green start beacon before trajectory execution.



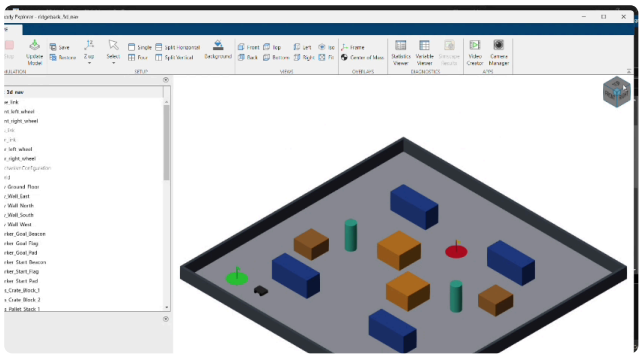
Step 2: South Corridor Acceleration

Chassis accelerates to 0.90 m/s heading toward the western turn.



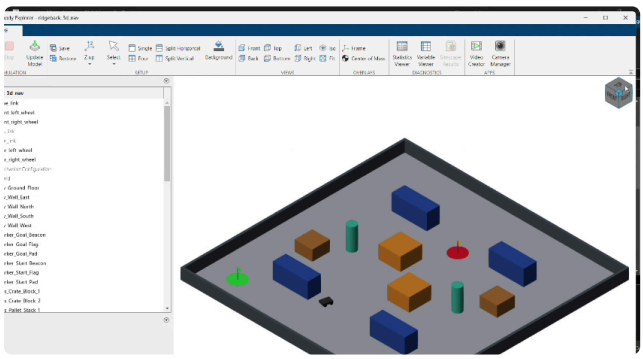
Step 3: Turning Around Rack SW

Executing smooth kinodynamic steering around warehouse shelving.



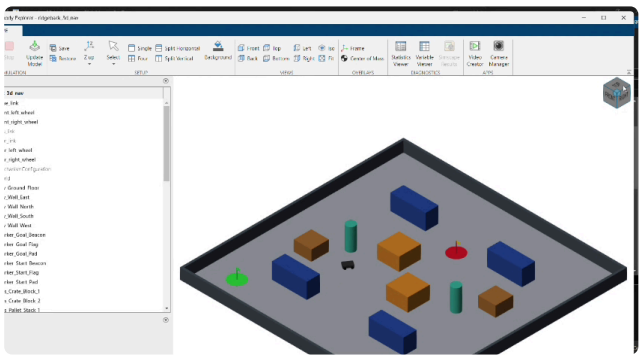
Step 4: Navigating Warehouse Aisle

Traversing the corridor between the shelving rack and central crates.



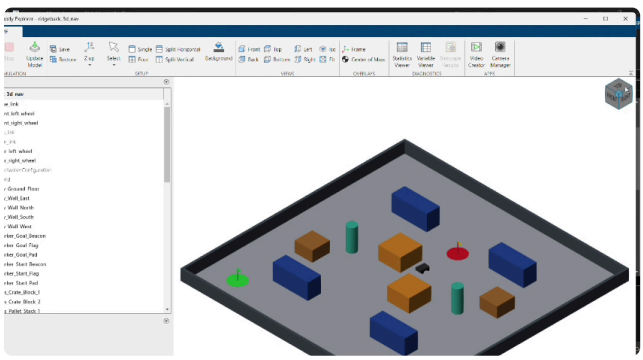
Step 5: Central Plaza Transit

Executing mid-course transit across open warehouse plaza.



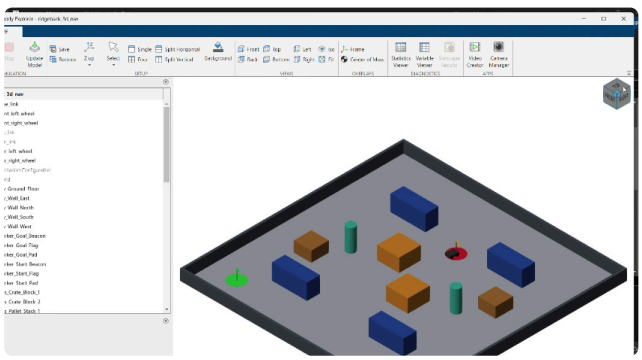
Step 6: Column Clearance & East Aisle

Passing structural column and aligning with the north-east approach.



Step 7: Target Approach & Deceleration

Cruising toward the red destination marker with smooth deceleration.



Step 8: Final Docking at Target [22.0, 22.0]

Autonomous halt centered directly over the target beacon pad.

7. Conclusion and Future Work

7.1 Summary of Contributions

This project successfully implemented an end-to-end, physics-accurate 3D autonomous navigation simulation for the Clearpath Ridgeback AMR in MATLAB and Simscape Multibody. Major milestones achieved include solving wheel-to-ground contact physics ($\Delta Z = +0.22$m), constructing an expanded 25m × 25m warehouse environment with 10 obstacle structures, deploying kinodynamic Hybrid A* global path planning and Pure Pursuit trajectory tracking, providing an interactive click-to-navigate GUI, configuring resilient model callbacks, and publishing the complete open-source codebase to GitHub.

7.2 Future Work

- **4-Wheel Independent Friction Contact:** Replace the prescribed planar joint with 4 independently actuated torque-controlled Mecanum wheel joints coupled with Simscape Spatial Contact Force blocks.
- **Dynamic Obstacle Avoidance:** Integrate local reactive planners (e.g., Timed Elastic Band or Dynamic Window Approach) to evade moving warehouse obstacles.
- **Perception and SLAM:** Simulate 3D LiDAR point clouds and stereo depth cameras to perform real-time SLAM and map generation.
- **ROS 2 / Nav2 Co-Simulation:** Interface the Simscape digital twin with ROS 2 Humble/Iron via the MATLAB ROS Toolbox for hardware-in-the-loop navigation benchmarking.

8. References

1. Clearpath Robotics, *Clearpath Ridgeback Mobile Robot User Manual and Specifications*, 2024.
2. MathWorks, *Simscape Multibody Reference Guide: Modeling Multibody Mechanical Systems*, MATLAB R2025b.
3. Dolgov, D., Thrun, S., Montemerlo, M., & Diebel, J., *Practical Search Techniques in Path Planning for Autonomous Driving*, AAAI Conference on Artificial Intelligence, 2008.
4. Coulter, R. C., *Implementation of the Pure Pursuit Path Tracking Algorithm*, Carnegie Mellon University Robotics Institute, CMU-RI-TR-92-01, 1992.